

Weather Data Analysis

Microsoft Excel FORMULAS & FUNCTIONS

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Dataset Overview

This dataset contains daily weather observations, including temperature, precipitation, date, and station information. It was cleaned and transformed to analyze weather patterns, classify climate conditions, identify freezing days, and prepare the data for reporting and dashboard visualization

Weather Data

Date	Month	Temperature_F	Precipitation_mm	Freeze	Climate	Precipitation Type	Conditions	Missing Data?
1-Jan-10	January	37	0	No	Cold	No Rain	Dry	FALSE
4-Jan-10	January	34	0	No	Cold	No Rain	Dry	FALSE
13-Jan-10	January	43	0	No	Cold	No Rain	Dry	FALSE
15-Jan-10	January	43	0	No	Cold	No Rain	Dry	FALSE
16-Jan-10	January	38	0	No	Cold	No Rain	Dry	FALSE
19-Jan-10	January	37	76	No	Cold	Heavy	Wet	FALSE
21-Jan-10	January	41	0	No	Cold	No Rain	Dry	FALSE
23-Jan-10	January	39	0	No	Cold	No Rain	Dry	FALSE
25-Jan-10	January	40	206	No	Cold	Heavy	Wet	FALSE
26-Jan-10	January	37	0	No	Cold	No Rain	Dry	FALSE
28-Jan-10	January	35	5	No	Cold	Light	Wet	FALSE
30-Jan-10	January	41	0	No	Cold	No Rain	Dry	FALSE
31-Jan-10	January	38	0	No	Cold	No Rain	Dry	FALSE
2-Feb-10	February	37	0	No	Cold	No Rain	Dry	FALSE
6-Feb-10	February	37	0	No	Cold	No Rain	Dry	FALSE
7-Feb-10	February	42	0	No	Cold	No Rain	Dry	FALSE
11-Feb-10	February	34	0	No	Cold	No Rain	Dry	FALSE

Weather Data Analysis Report

This project analyzes historical weather data from **Boston Logan Airport**. The dataset was cleaned by removing missing and invalid values, converting the date into a proper Excel format, and creating new calculated columns such as **Freeze**, **Climate**, **Precipitation Type**, **Conditions**, and **Missing Data**. These transformations made the dataset accurate, consistent, and ready for analysis.

The cleaned data was used to identify weather patterns, including freezing days, rainfall intensity, and wet versus dry conditions. This project demonstrates practical Excel skills in **data cleaning**, **transformation**, **logical functions (IF/IFS)**, **Excel Tables**, and **data preparation**.